

Introverts vs. Extroverts

A Visualization

Section 1. Introduction

This project dives into the behavioral patterns of introverts and extroverts. By asking questions and creating visuals to help answer these questions, these two personalities can be easily compared. Through a mix of bar charts, a scatter plot, a box plot, and even a radar plot, the public can compare introverts and extroverts.

Tableau Public Link:

https://public.tableau.com/app/profile/daniel.miller7456/viz/Miller_A7_Final/Goal1

Section 2. Data Description

- Data Domain
 - o Behavioral patterns of introverts and extroverts
- Data file
 - o A CSV file with data structured in a tabular format
- Rows & Columns
 - o The dataset has 2,900 rows and this project used 15 of 21 total columns
 - o The dataset originally only had 8 columns; feature engineering was done to create more.
- Data Source
 - o This dataset was found on Kaggle and was created by Rakesh Kapilavai.
 - o Data Source Link: <https://www.kaggle.com/datasets/rakeshkapilavai/extrovert-vs-introvert-behavior-data>

Section 3. Data Cleaning Strategies

The data did not need to be cleaned, but as stated before there was some feature engineering done.

Numerous columns were scaled to be valued between 0 and 1.

Other new columns were created by taking a difference calculation as one of the charts is a sorted bar chart of the average difference of behavioral patterns between introverts and extroverts.

Online and offline score columns were calculated by compiling available columns. For example, combining social_event_attendance, going_outside, friends_circle_size, and subtracting time_spent_alone to create the offline score column.

A social score column was calculated in a similar way, instead of just online or offline the social score took both online and offline scores and combined them.

Section 4. Clean Dataset

The columns of the full dataset are as follows:

- Time spent Alone
 - Stage fear
 - Social event attendance
 - Going outside
 - Drained after socializing
 - Friends circle size
 - Post frequency
 - Personality
 - Social score
 - Scaled social score
 - Scaled post frequency/Scaled online score
 - Offline score
 - Scaled offline score
 - Behavior
 - Score
 - Social Score Category
 - Diff friends circle size
 - Diff time spent alone
 - Diff going outside
 - Diff post frequency
 - Diff social event attendance
-
- = Pivot created in Tableau
 - = Calculation made in Tableau

Data Excerpt:

# personality/dataset	Abc personality/dataset	# personality/dataset	# personality/dataset	Abc personality/dataset	# personality/dataset	# personality/dataset	Abc personality/dataset
Time spent Alone	Stage fear	Social event attendance	Going outside	Drained after socializing	Friends circle size	Post frequency	Personality
4.0000	No	4.00000	6	No	13.0000	5	Extrovert
4.0000	No	4.00000	6	No	13.0000	5	Extrovert
4.0000	No	4.00000	6	No	13.0000	5	Extrovert
4.0000	No	4.00000	6	No	13.0000	5	Extrovert
9.0000	Yes	0.00000	0	Yes	0.0000	3	Introvert
9.0000	Yes	0.00000	0	Yes	0.0000	3	Introvert
9.0000	Yes	0.00000	0	Yes	0.0000	3	Introvert
9.0000	Yes	0.00000	0	Yes	0.0000	3	Introvert
9.0000	Yes	1.00000	2	Yes	5.0000	2	Introvert
9.0000	Yes	1.00000	2	Yes	5.0000	2	Introvert
9.0000	Yes	1.00000	2	Yes	5.0000	2	Introvert
9.0000	Yes	1.00000	2	Yes	5.0000	2	Introvert
0.0000	No	6.00000	7	No	14.0000	8	Extrovert
0.0000	No	6.00000	7	No	14.0000	8	Extrovert
0.0000	No	6.00000	7	No	14.0000	8	Extrovert
0.0000	No	6.00000	7	No	14.0000	8	Extrovert

# personality/dataset	# personality/dataset	# personality/dataset	# personality/dataset	# personality/dataset	Abc Pivot	# Pivot	Abc Calculation
Social score	Scaled social score	Scaled online score	Offline score	Scaled offline score	Behavior	Score	Social Score Category
1.76017	0.720045	0.50000	2.12381	0.749580	Scaled friends circle size	0.86667	High
1.76017	0.720045	0.50000	2.12381	0.749580	Scaled going outside	0.85714	High
1.76017	0.720045	0.50000	2.12381	0.749580	Scaled social event attendance	0.40000	High
1.76017	0.720045	0.50000	2.12381	0.749580	Scaled time spent alone	0.36364	High
-0.81818	0.047431	0.30000	0.00000	0.000000	Scaled friends circle size	0.00000	Low
-0.81818	0.047431	0.30000	0.00000	0.000000	Scaled going outside	0.00000	Low
-0.81818	0.047431	0.30000	0.00000	0.000000	Scaled social event attendance	0.00000	Low
-0.81818	0.047431	0.30000	0.00000	0.000000	Scaled time spent alone	0.81818	Low
-0.09913	0.235008	0.20000	0.71905	0.253782	Scaled friends circle size	0.33333	Low
-0.09913	0.235008	0.20000	0.71905	0.253782	Scaled going outside	0.28571	Low
-0.09913	0.235008	0.20000	0.71905	0.253782	Scaled social event attendance	0.10000	Low
-0.09913	0.235008	0.20000	0.71905	0.253782	Scaled time spent alone	0.81818	Low
2.53333	0.921739	0.80000	2.53333	0.894118	Scaled friends circle size	0.93333	High
2.53333	0.921739	0.80000	2.53333	0.894118	Scaled going outside	1.00000	High
2.53333	0.921739	0.80000	2.53333	0.894118	Scaled social event attendance	0.60000	High
2.53333	0.921739	0.80000	2.53333	0.894118	Scaled time spent alone	0.00000	High

The Diff columns are not in these images because they only hold null values, as they are the average difference between introverts and extroverts of a single column, for example time spent alone. They are one value, not an entire column, even though Tableau makes them a column.

Section 5. Visualization Tools

Excel & Tableau were the software used to create this project, with Excel's usage being exclusively for feature engineering while Tableau was used for future engineering and creating visualizations.

The tools in this project were bar charts, a scatter plot, box plot, and a radar chart.

There were three different bar charts, and two of them show a distribution of a whole while the third (sorted chart) shows how similar/different certain behaviors are between the two personalities.

The scatter plot was chosen to show where introverts and extroverts lie in terms of online and offline engagement. With one axis for socializing online and the other offline. When sorted by color, the scatter shows the relationship between the two variables.

Box plots are great for showing distribution and spread of a numerical variable. In this case, the variable is time spent alone.

The radar chart was chosen for its ability to compare multiple behavioral patterns for both personalities all on one chart.

Link to Charts:

https://public.tableau.com/app/profile/daniel.miller7456/viz/Miller_A7_Final/Goal1

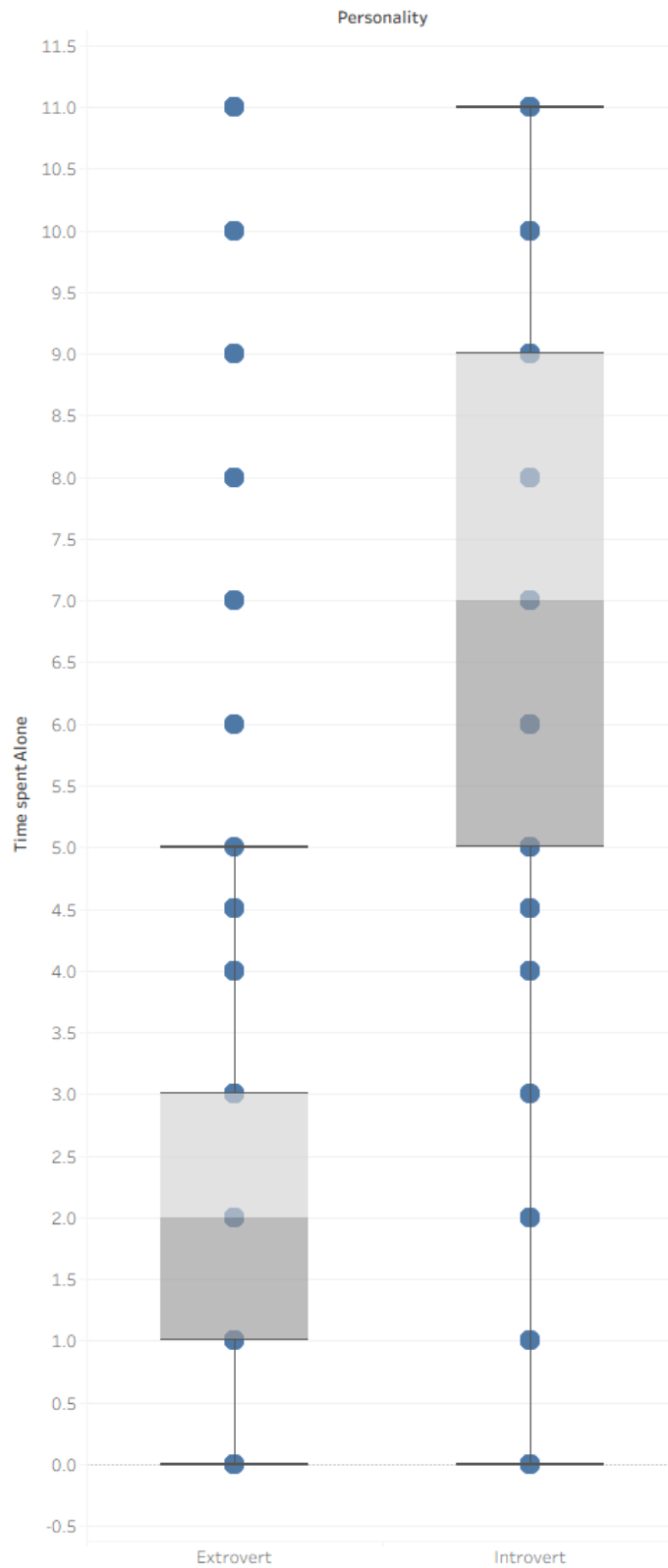
Section 6. Visualizations and Stories

Six goals were thought up for this project and a chart and story were produced for each one.

Goal 1.

Goal – Does one personality spend more time alone than the other?

Chart – Does One Personality Spend More Time Alone



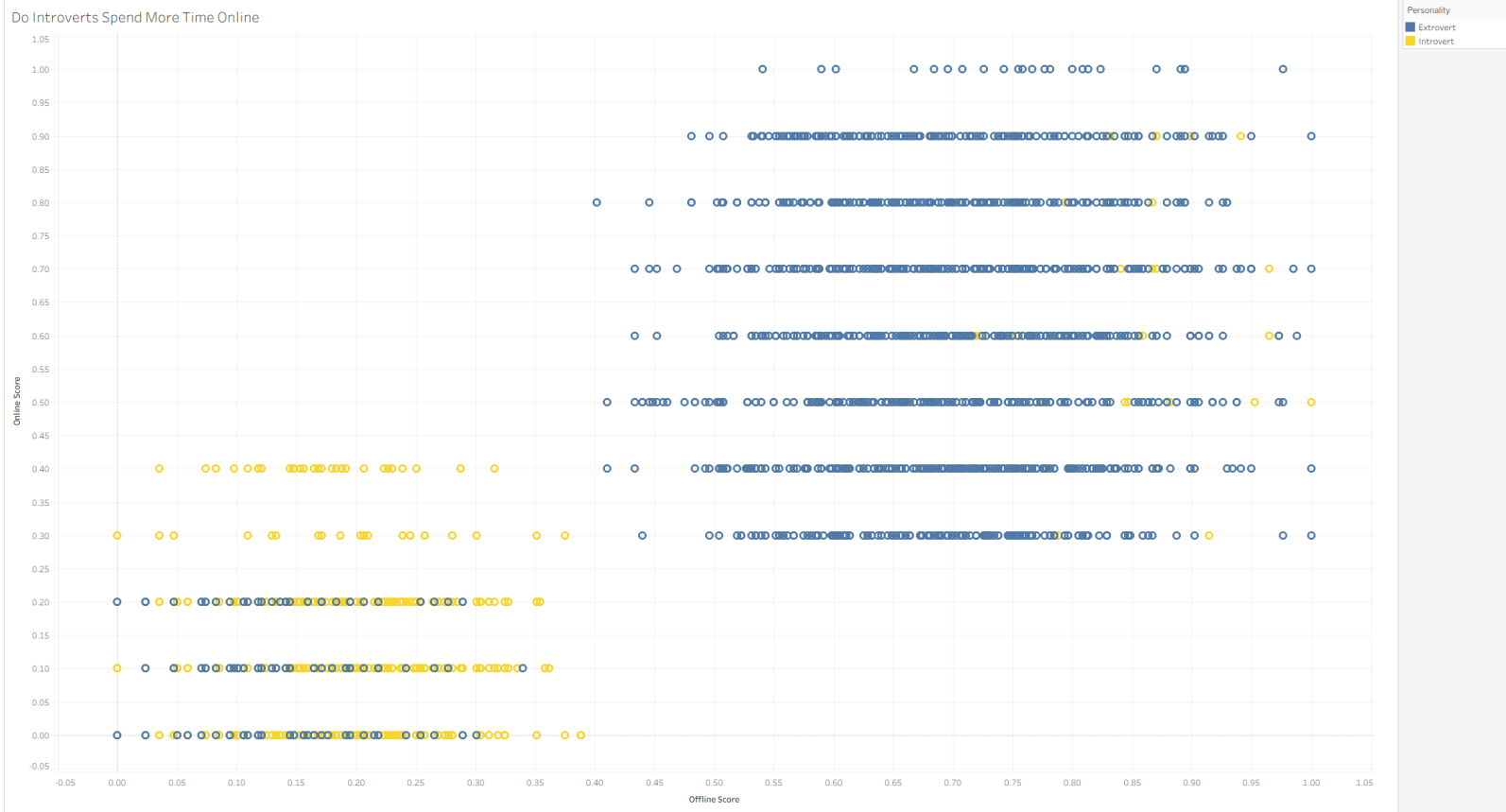
Story –

Extroverts' maximum time spent alone is the same value as the introverts' lower quartile. This means that every extrovert spends less time alone than 75% of introverts. While both personalities have the same minimum time, their maximums and medians distinctly vary, with the extroverts' respective values being 5 and 2 while introverts' are 11 and 7, showing just how much more time introverts spend alone.

Goal 2.

Goal – Do introverts spend more time engaging online while extroverts spend more time engaging offline?

Chart –



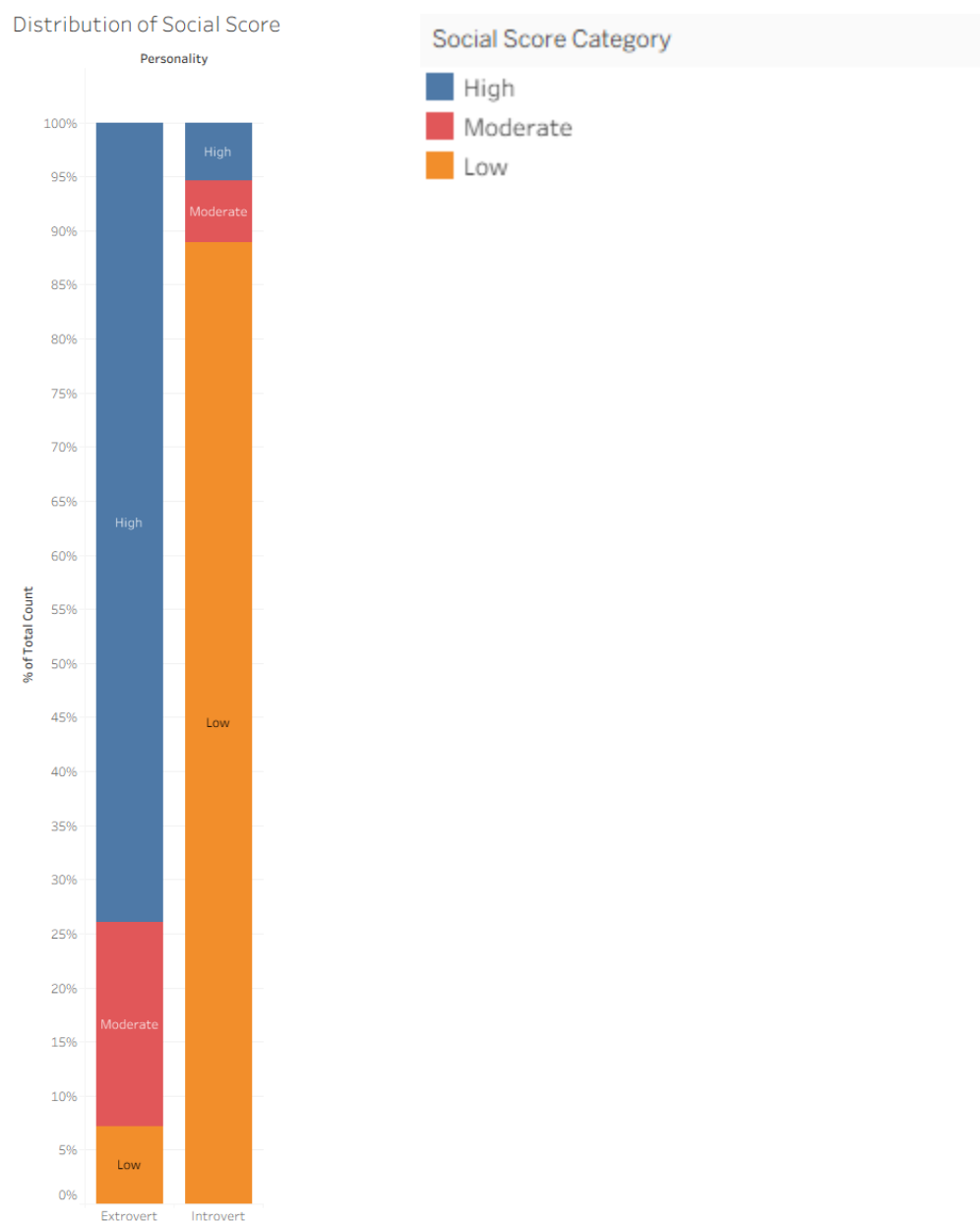
Story –

The scatter plot shows two big groups: those with both low offline and low online scores and those with both high offline and high online scores. There is also a group of those with high offline scores and low online scores, and nearly all individuals in this group are extroverts. The two big groups created also show a clear divide in personality, as there are a few outliers here and there, but for the most part, those with low scores are introverts while those with high scores are extroverts. This chart shows that introverts do not spend more time engaging online.

Goal 3.

Goal – What is the distribution of each personality's social score?

Chart –



Story –

Nearly 90% of introverts fall into the low social score category while only 7% of extroverts are in the same category. In both personalities, moderately social makes up a small amount (~18% extroverts and 5% introverts). The majority of extroverts, almost 75%, fall into the high social score category while only about 5% of introverts are in the same category. Extroverts are much more likely to be highly social while introverts are much more likely to be lowly social.

Goal 4.

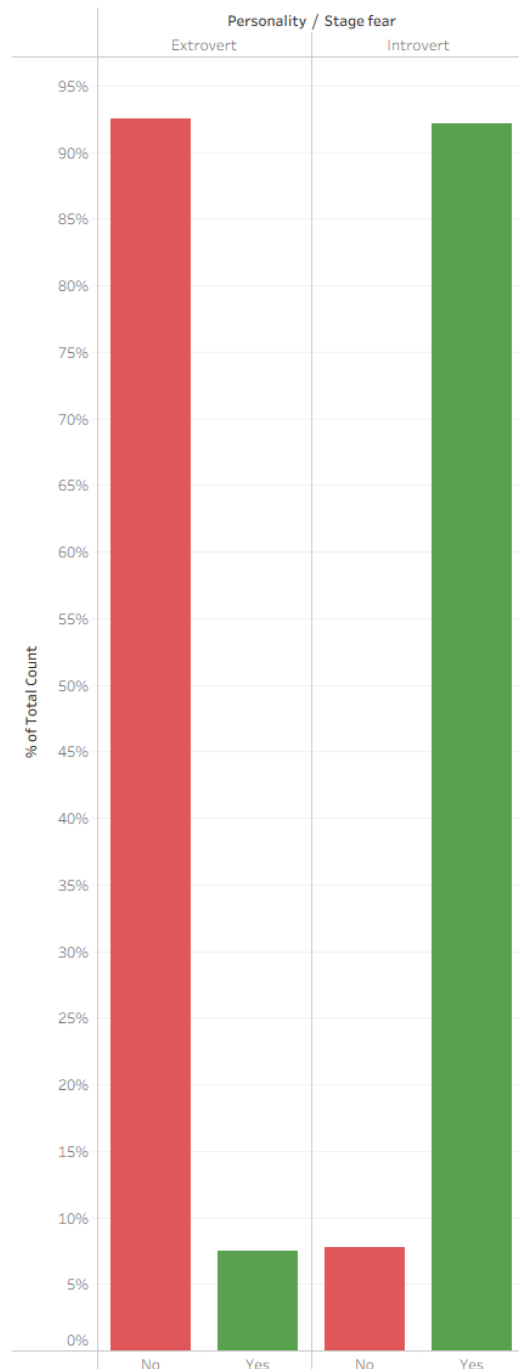
Goal – Are extroverts less likely to have stage fright?

Chart –

Story –

With almost mirrored bars, introverts and extroverts are just about polar opposites in terms of having stage fright. While roughly 7% of extroverts say they feel stage fright, a similar 7% of introverts say they don't feel stage fright. This means that the majority of extroverts don't have stage fright and the majority of introverts have stage fright, showcasing the clear divide between the two personalities.

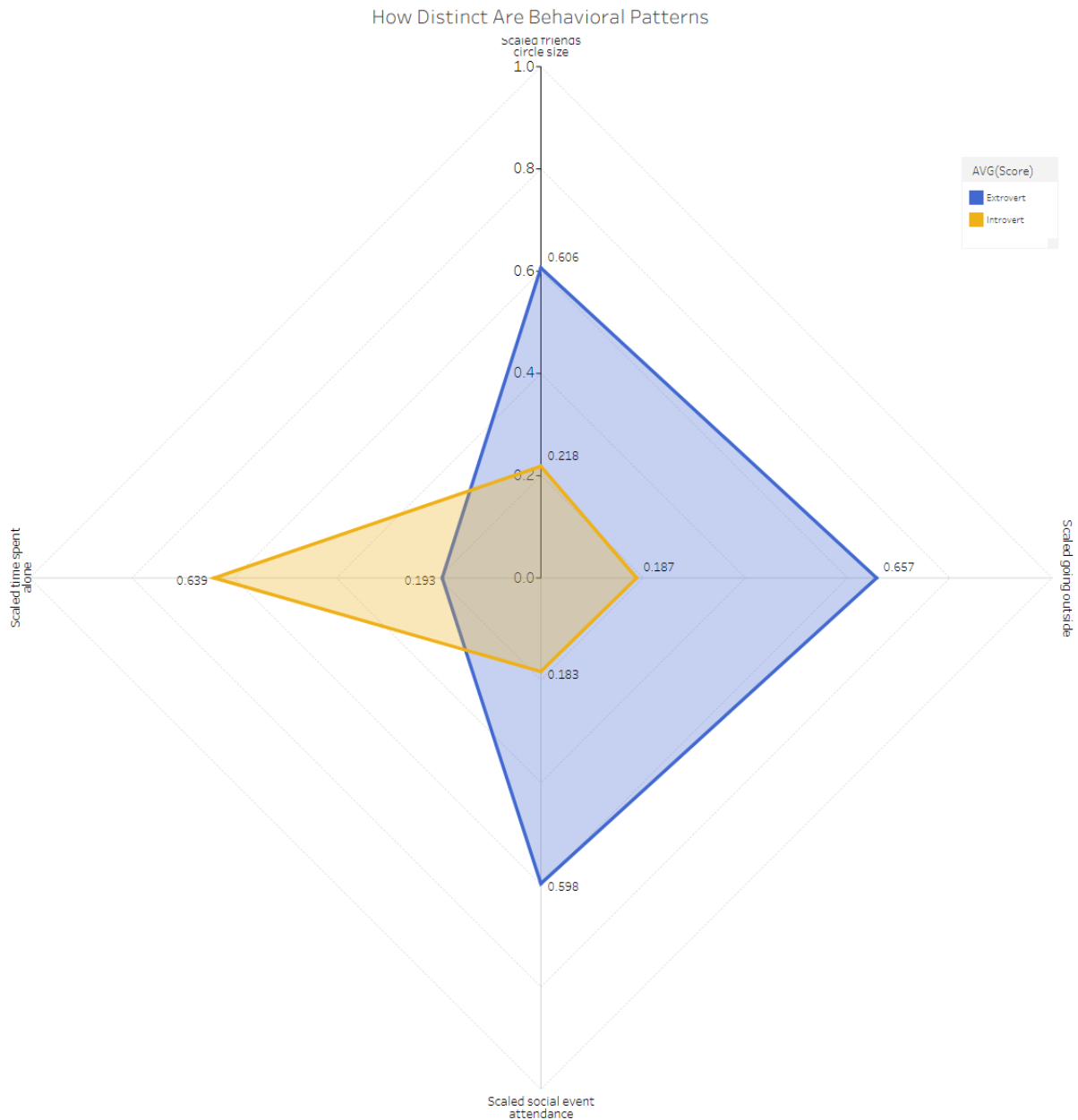
Are Extroverts Less Likely to Have Stage Fright



Goal 5.

Goal – How distinct are behavioral patterns between the personality types?

Chart –



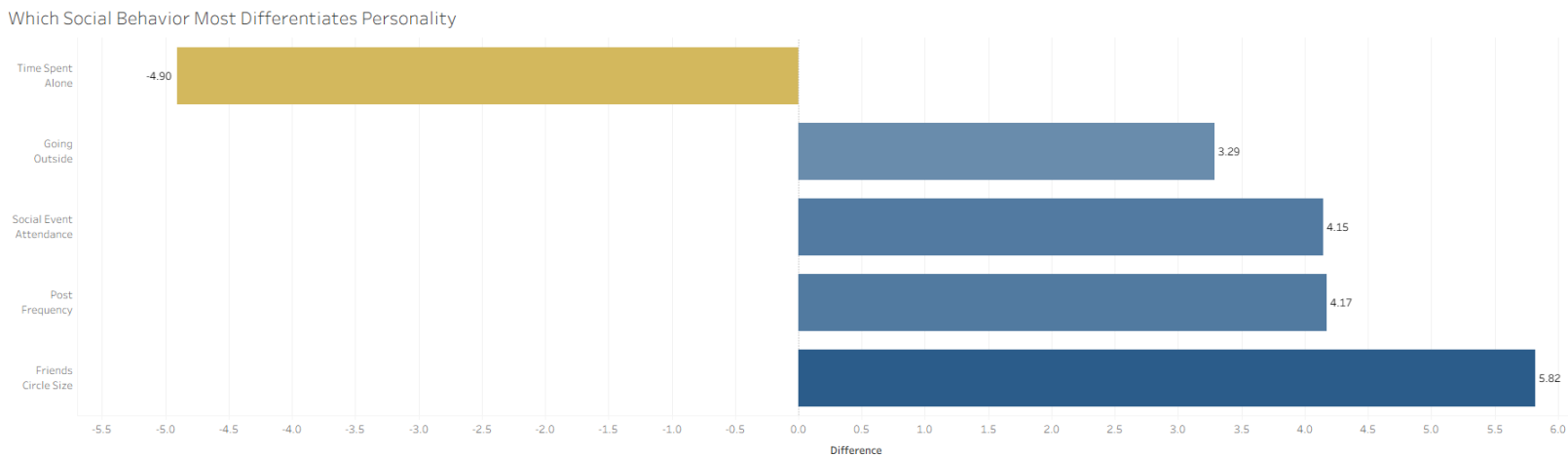
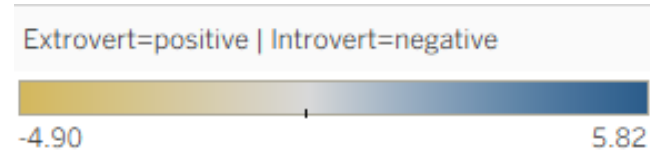
Story –

This radar chart shows the average score of four metrics for both introverts and extroverts. Every metric was scaled to be between the values of 0 and 1. Extroverts dominate when discussing friends circle size, time spent outside, and the number of social events attended. Meanwhile, introverts dominate when talking about time spent alone. Neither personality reaches out to the 1 point, which showcases the variety within each personality type.

Goal 6.

Goal – Which social behavior most differentiates personality?

Chart –



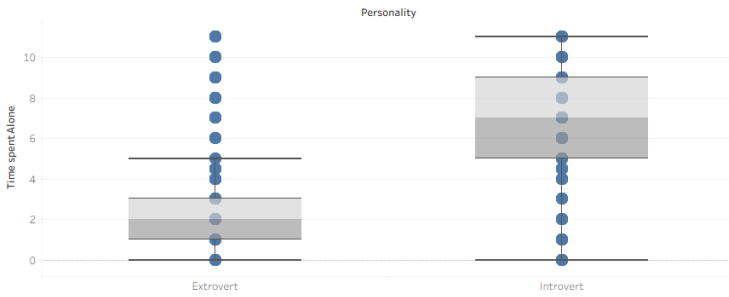
Story –

Friends circle size is the behavioral pattern that most differentiates the two personality types. This chart shows the average difference in values between introverts and extroverts. The more positive the difference, the more extrovert leaning that behavior is, and vice versa. The only behavior that leans towards introverts is time spent alone, which has proven, through previous charts, to be a very strong introvert behavior. Going outside, social event attendance, and post frequency all have similar differences, and all three behaviors are extrovert leaning.

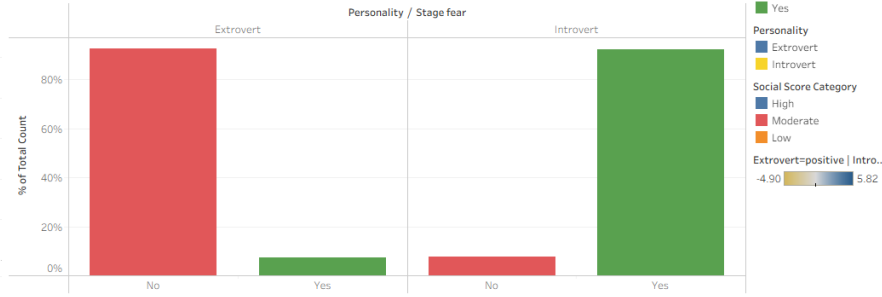
Final Dashboard

Chart —

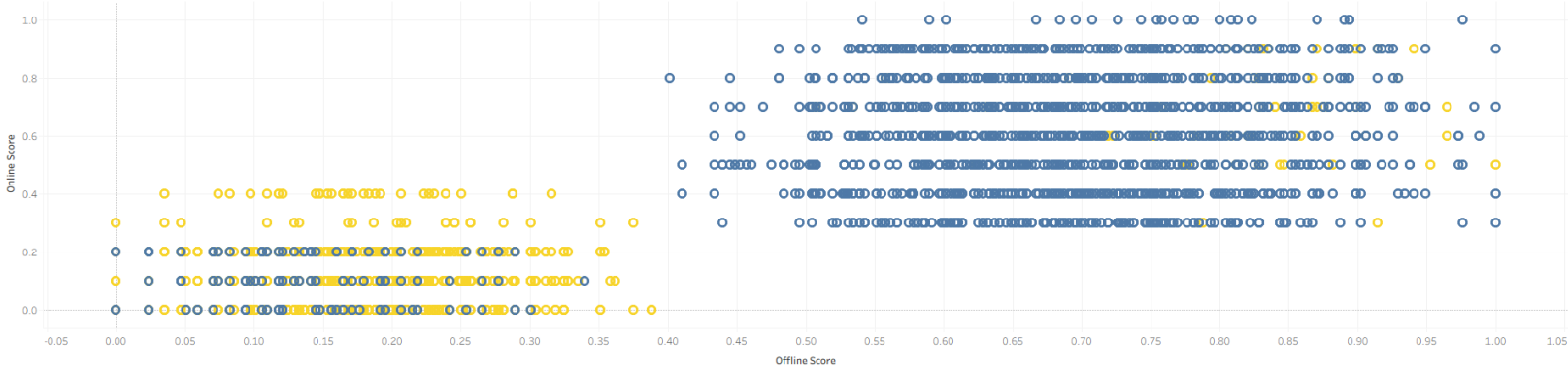
Does One Personality Spend More Time Alone



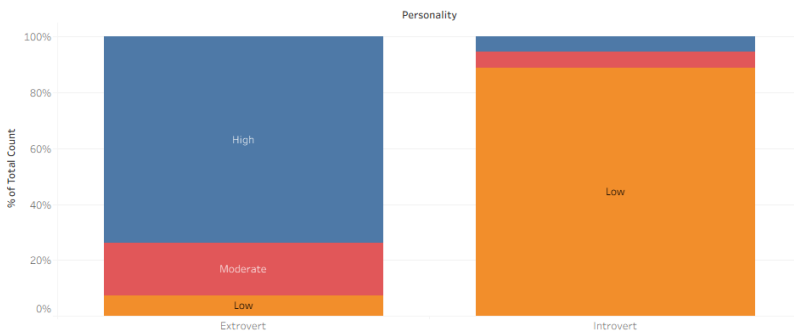
Are Extroverts Less Likely to Have Stage Fright



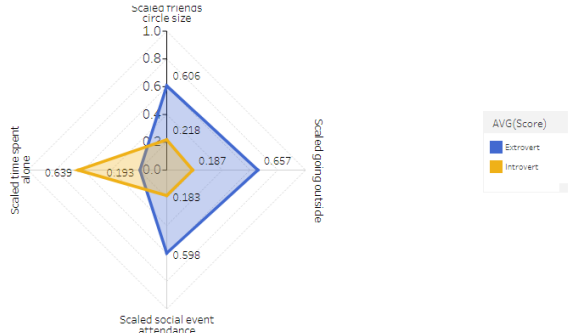
Do Introverts Spend More Time Online



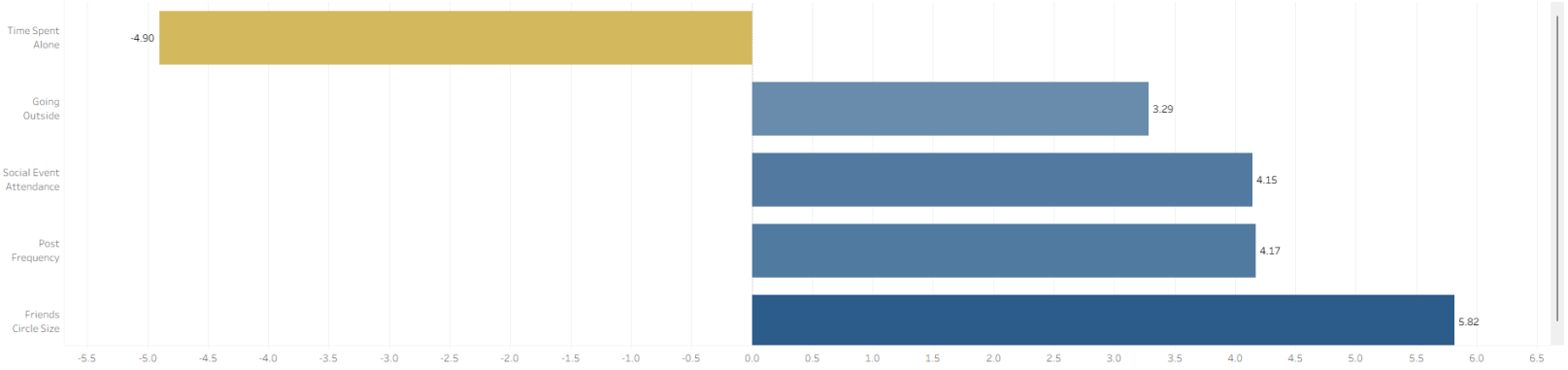
Distribution of Social Score



How Distinct Are Behavioral Patterns



Which Social Behavior Most Differentiates Personality



Story –

With every chart together, the clear divide between introverts and extroverts becomes apparent. These two personalities differ for nearly every single metric. Whether it's time spent alone or stage fright, these personalities are opposites. The two groups in the scatter plot also show how divided they can be. The radar chart and sorted bar both show how much time introverts choose to spend alone, which helps prove why most introverts in the stacked bar chart are in the low social category. Ultimately, this dashboard clearly illustrates how everyday behaviors form distinct patterns that separate introverts and extroverts.

Section 7. Conclusions

Overall, the analysis confirmed that introverts and extroverts are distinct personalities with little in common. Introverts spend much more time alone with a low social score. Even with all that time alone, they barely spend it socializing online. Extroverts on the other hand are socializing online just as much as they are offline. They also have a much higher social score, as their average friend group size, outside time, and social event attendance are much higher than the averages of introverts. In most cases, the extroverts' averages are over triple that of the introverts. This dataset would be of great use to predictable machine learning. To estimate whether someone is an introvert or extrovert can be done.